

Lesson 5

Quadratic forms in normal vectors, and the F test of $R\beta = r$

1 The model and the hypothesis (a reminder)

The set opens with the model of the previous lessons, now with normal errors,

$$y = X\beta + u, \quad u \sim N(0, \sigma^2 I),$$

and with the linear hypothesis on which Lesson 4 ended,

$$H_0 : \underbrace{R}_{q \times K} \underbrace{\beta}_{K \times 1} = \underbrace{r}_{q \times 1}.$$

To test it, the lecture first needs the distribution of quadratic forms in a normal vector; it takes a digression to get them.

2 Digression: quadratic forms and the χ^2 distribution

(1) A standard normal vector

Let $\underbrace{z}_{n \times 1} \sim N(0, I_n)$. Then

$$\sum_{i=1}^n z_i^2 = \underbrace{z'}_{1 \times n} \underbrace{z}_{n \times 1} \sim \chi_n^2,$$

a sum of n independent squared standard normals.

(2) A normal vector with a general covariance

If $\underbrace{x}_{n \times 1} \sim N(0, \Sigma)$, with Σ an $n \times n$ (positive definite) covariance matrix, then

$$x' \Sigma^{-1} x \sim \chi_n^2.$$

The proof runs through the spectral decomposition.

Spectral decomposition theorem

Let Σ be symmetric and real. Then it has eigenvalues and eigenvectors,

$$\underbrace{\Sigma}_{n \times n} \underbrace{q}_{n \times 1} = \underbrace{\lambda}_{\text{scalar}} q,$$

and collecting the eigenvectors in the columns of Q ,

$$Q' \Sigma Q = \Lambda, \quad \Lambda = \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix}, \quad \text{where } Q' = Q^{-1} \text{ (an orthogonal matrix).}$$

Multiplying by Q on the left and Q' on the right,

$$QQ'\Sigma QQ' = Q\Lambda Q' \implies \Sigma = Q\Lambda Q' = Q\Lambda^{1/2}\Lambda^{1/2}Q', \quad \Lambda^{1/2} = \begin{bmatrix} \sqrt{\lambda_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sqrt{\lambda_n} \end{bmatrix},$$

so that, with $T = Q\Lambda^{1/2}$,

$$TT' = Q\Lambda^{1/2}\Lambda^{1/2}Q' = \Sigma.$$

The proof of (2)

Take $\underbrace{z}_{n \times 1} \sim N(0, I_n)$ and set $x = Tz$. Then

$$\mathbb{E}(x) = T\mathbb{E}(z) = 0, \quad \text{Var}(x) = T\text{Var}(z)T' = TT' = \Sigma,$$

so $x \sim N(0, \Sigma)$. Conversely $z = T^{-1}x$, and $z'z \sim \chi_n^2$ by (1), so

$$z'z = x'(T^{-1})'T^{-1}x = x'(T')^{-1}T^{-1}x = x'\Sigma^{-1}x \sim \chi_n^2,$$

using

$$\Sigma^{-1} = (TT')^{-1} = (T')^{-1}(T)^{-1}.$$

(3) An idempotent quadratic form

Let $z \sim N(0, I_n)$ and let A be (symmetric and) idempotent of rank p . Then

$$z'Az \sim \chi_p^2.$$

First, the eigenvalues of an idempotent matrix: from $AA = A$ and $Aq = \lambda q$,

$$A(Aq) = Aq = \lambda q \quad \text{and} \quad A(Aq) = A\lambda q = \lambda Aq = \lambda^2 q,$$

so

$$\lambda q = \lambda^2 q \implies (\lambda^2 - \lambda)q = 0 \implies \lambda = 1 \text{ or } \lambda = 0.$$

Hence the rank equals the trace:

$$\text{rank}(A) = \text{trace}(A),$$

because with $Q'AQ = \Lambda$,

$$\text{trace}(Q'AQ) = \text{trace}(\Lambda) = \sum_{i=1}^n \lambda_i = p = \text{trace}(A),$$

the trace being invariant ($\text{trace}(Q'AQ) = \text{trace}(AQQ') = \text{trace}(A)$) and the number of unit eigenvalues being the rank.

Now rotate z : let $z = Qy$, i.e. $y = Q'z$. Then

$$\mathbb{E}(y) = 0, \quad \text{Var}(y) = Q'\text{Var}(z)Q = Q'Q = I_n,$$

so $y \sim N(0, I_n)$ and $y'y \sim \chi_n^2$. The quadratic form becomes

$$z'Az = y'\underbrace{Q'AQ}_{\Lambda}y = y'\Lambda y = \sum_{i=1}^p y_i^2 \sim \chi_p^2,$$

ordering the eigenvalues so that the p ones come first.

(4) Independence of two quadratic forms

Let Az and Bz be two normal vectors, where A and B are idempotent and symmetric, of ranks p and q respectively. By (3),

$$w_1 = z'Az \sim \chi_p^2, \quad w_2 = z'Bz \sim \chi_q^2,$$

and

$$w_1 \text{ is independent of } w_2 \iff AB = 0.$$

The reason: $w_1 = (Az)'(Az)$ depends on z only through Az , and w_2 only through Bz ; these two normal vectors have covariance

$$\mathbb{E}(Az(Bz)') = A \text{Var}(z)B' = AB = 0,$$

and uncorrelated jointly normal vectors are independent.

Definition (F distribution)

Let $w_1 \sim \chi_p^2$ and $w_2 \sim \chi_q^2$ be independent. Then

$$\frac{w_1/p}{w_2/q} \sim F(p, q).$$

In particular, if $w_1 \sim \chi_1^2$,

$$\frac{w_1/1}{w_2/q} \sim F(1, q) = t_q^2.$$

3 The test of $H_0 : R\beta = r$

The distance between Rb and r

The hypothesis is judged by how far Rb lies from r , written $|Rb - r|$ and measured, by (2), as a quadratic form weighted by the inverse covariance — the $x'\Sigma^{-1}x$ written beside it on the page:

$$|Rb - r| \longrightarrow (Rb - r)' [\text{Var}(Rb - r)]^{-1} (Rb - r) \sim \chi_q^2.$$

$Rb - r$ under the null

$$\begin{aligned} Rb - r &= R(X'X)^{-1}X'y - r \\ &= R\beta + R(X'X)^{-1}X'u - r \\ &= R(X'X)^{-1}X'u \quad \text{when } H_0 \text{ holds.} \end{aligned}$$

Its variance

Using $\text{Var}(Ax) = A \text{Var}(x)A'$,

$$\begin{aligned} \text{Var}(Rb - r) &= \text{Var}(R(X'X)^{-1}X'u) \\ &= R(X'X)^{-1}X' \underbrace{\mathbb{E}(uu')}_{\sigma^2 I} X(X'X)^{-1}R' \\ &= \sigma^2 (R(X'X)^{-1}R'). \end{aligned}$$

The quadratic form (the “sandwich”)

The inverse variance is sandwiched between $(Rb - r)'$ and $(Rb - r)$:

$$|Rb - r| = (Rb - r)' [\text{Var}(Rb - r)]^{-1} (Rb - r), \quad Rb - r = R(X'X)^{-1}X'u,$$

so

$$|Rb - r| = \frac{1}{\sigma^2} u' \underbrace{X(X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}R(X'X)^{-1}X'}_N u = \frac{1}{\sigma^2} u'Nu.$$

The properties of N

$$N' = N, \quad NN = N, \quad MN = MX(\cdots) = 0,$$

and its rank is its trace,

$$\begin{aligned} \text{rank}(N) &= \text{trace}(N) = \text{trace} \left[(X(X'X)^{-1}R') (R(X'X)^{-1}R')^{-1} R(X'X)^{-1}X' \right] \\ &= \text{trace} \left[(R(X'X)^{-1}R')^{-1} (R(X'X)^{-1}X'X(X'X)^{-1}R') \right] \\ &= \text{trace}(I_q) = q, \end{aligned}$$

the middle step moving the first factor to the back (the trace is invariant under cyclic permutation) and the last using $X'X(X'X)^{-1} = I$.

The two χ^2 pieces

The unknown σ^2 is estimated by

$$s^2 = \frac{u'Mu}{n - K} \quad (= e'e/(n - K)),$$

and, since $u/\sigma \sim N(0, I_n)$, result (3) gives

$$\textcircled{1} \quad \frac{1}{\sigma^2} u'Nu \sim \chi_q^2, \quad \textcircled{2} \quad \frac{1}{\sigma^2} u'Mu \sim \chi_{n-K}^2,$$

and they are independent by (4), because

$$MN = 0 \quad \text{because} \quad MX = 0.$$

The F statistic

$$\frac{\frac{1}{\sigma^2} u'Nu / q}{\frac{1}{\sigma^2} u'Mu / (n - K)} \sim F(q, n - K),$$

the numerator being the χ_q^2 and the denominator the χ_{n-K}^2 , each divided by its degrees of freedom. The σ^2 cancels, so the statistic can be computed from the data:

$$F = \frac{(Rb - r)' [R(X'X)^{-1}R']^{-1} (Rb - r) / q}{s^2} \sim F(q, n - K) \quad \text{under } H_0.$$

Transcription notes

- The set opens with a divider page carrying only “Lesson 5”; it is not reproduced.
- Transposes are written both as T and as $'$ in the original; all are typeset $'$, as in Lesson 4. The regressor matrix is written lower case (x) on the page and typeset X , as in Lessons 3–4; in the digression the lower-case x is a random vector and is kept lower case. The page writes B for β in $R\beta = r$ and in $R\beta + R(X'X)^{-1}X'u - r$; typeset β . The error variance, written with a glyph like 6, is σ^2 . The page’s K/k are both typeset K .
- The digression numbers only items (3) and (4); the headings (1) and (2) are **added** for the two results that precede them on the page ($z'z \sim \chi_n^2$ and $x'\Sigma^{-1}x \sim \chi_n^2$).
- In (2) the page writes $N(0, \Sigma)$ with an n under Σ , read as its dimension; “positive definite” is **added** (needed for Σ^{-1} and $\sqrt{\lambda_i}$). $T = Q\Lambda^{1/2}$ is not named on the page, which goes from $Q\Lambda^{1/2}\Lambda^{1/2}Q' = \Sigma$ straight to $TT' = \Sigma$; it is **added**.
- **Corrected:** in the proof of (2) the page writes $x'(T^{-1})'T^{-1}z = x'(X')^{-1}T^{-1}x$; the first line should end in x and the middle factor is $(T')^{-1}$. The page’s “ $\Sigma^{-1} = (T')^{-1}(T)^{-1}$ ” is kept, with $(TT')^{-1}$ added in front.
- **Corrected:** in (3) the eigenvalue step is written “ $A(Aq) = A\lambda q$, $Aq = \lambda Aq = \lambda^2 q$ ”; typeset as the two evaluations of $A(Aq)$ it intends. (3) says only “idempotent”; “symmetric” is added in brackets, since the spectral decomposition and the orthogonal Q need it (and (4) states it).
- **Corrected:** “ $\text{Var}(y) = Q' \text{Var}(z) = Q = I_n$ ” on page 4 drops the right-hand Q ; typeset $Q' \text{Var}(z)Q = Q'Q = I_n$. The page draws an arrow labelled Λ at $Q'AQ$; typeset as an underbrace. The subscript of $y'y \sim \chi^2$ is unclear; it is n . The trace invariance and the ordering of the eigenvalues are **added**. **Corrected:** the page writes $z'Az = y'Q'AQ'y$; since $z = Qy$ the last factor is Q , giving $y'Q'AQy$.
- (4) begins “Let Ax and Bx be two normal vector”; the vector is z in everything that follows, so it is typeset Az , Bz . **Corrected:** the covariance is written $A \text{Var}(z)\beta = AB$; the last factor is B' ($= B$ by symmetry). The sentence explaining why zero covariance gives independence is **added**. Note that the “iff” is as written on the page.
- **Corrected:** on page 5 the quadratic form is written $(Rb - r)'(\text{Var}(Rb - r))(Rb - r)$ without the inverse; page 6 writes it with the inverse, which is right (and matches $x'\Sigma^{-1}x$ beside it).
- **Corrected:** the variance line on page 6 is written $R(X'X)^{-1}\mathbb{E}(uu')X(X'X)^{-1}R'$, dropping the X' before $\mathbb{E}(uu')$.
- **Corrected:** the expansion of $|Rb - r|$ is written $\sigma^2 u'X(X'X)^{-1}R(R(X'X)^{-1}R')^{-1}R(X'X)^{-1}X'u$; the constant is $1/\sigma^2$ (the inverse of $\sigma^2 R(X'X)^{-1}R'$), and the first R carries a prime, as in the definition of N in the trace line on page 7. The $1/\sigma^2$ reappears correctly on page 8.
- **Corrected:** in the second trace line on page 7 the first bracket is written $(R(X'X)^{-1}R)^{-1}$ and the second contains $X'Y$; they are $(R(X'X)^{-1}R')^{-1}$ and $X'X$. “ $mN = MX(\dots)$ ” is written without a right-hand side; $= 0$ is added (page 8 states $MN = 0$).
- **Corrected:** page 8 writes ① as $u'Nu \sim \chi_q^2$ and ② as $u'Mu \sim \chi_{n-K}^2$ without the $1/\sigma^2$; the factor is inserted (the F ratio just below carries it).
- **Added:** $s^2 = e'e/(n - K)$ beside $u'Mu/(n - K)$, and the closing formula for F with σ^2 cancelled — the page stops at $F(q, n - K)$ drawn with arrows from the two χ^2 pieces.
- Pages of the original, for tracing back: page 1 the divider; page 2 the model, H_0 with dimensions, the digression heading, (1), (2), and the spectral decomposition up to $\Lambda^{1/2}$; page 3 $TT' = \Sigma$, the proof of (2), and the statement of (3) with the eigenvalue step; page 4 $\lambda = 1$ or 0 , rank = trace, the rotation $y = Q'z$ and $z'Az \sim \chi_p^2$, and the statement of (4); page 5 w_1, w_2 , the independence condition, the definition of F , $F(1, q) = t_q^2$, H_0 and $|Rb - r|$, and the expansion of $Rb - r$; page 6 $Rb - r$ under H_0 , its variance (with $\text{Var}(Ax) = A \text{Var}(x)A'$), the quadratic form (labelled “sandwich”, with an arrow to it) and the underbraced N ; page 7 $u'Nu$, MN , the properties of N , its trace, and s^2 ; page 8 the two χ^2 pieces, $MN = 0$, and the

F ratio.